

Working title: Parameter efficient fine-tuning ensembles in IMDB- reviews

Problem description

This project is about fine tuning pre-existing Large-Language models such as RoBERTa or DeBERTa utilizing LoRA combined with ensemble methods. It will be utilized to automatically categorize IMDB movie reviews into positive and negative reviews to create a snapshot of the sentiment or mood of the audience. As the movie industry is a multi-billion dollar business, it is very important to know how a certain movie is received by the audience, considering the revenue. In the architecture we will be using, RoBERTa and DeBERTa can use a maximum of 512 tokens. The comments within the IMDB dataset contain data that exceeds this limit; therefore, if the data cannot be retrieved from the beginning or end, an incorrect conclusion may be reached. Data Leakage and Overfitting Risk in Stacking: If the Level 1 Meta-Learner is trained and validated using the same data we used to train our Level 0 models, and the resulting probability estimates are used again, an overfitting scenario will likely occur. To resolve this, we will adhere to the Out-of-Fold (OOF) production of potential sequels or merchandise articles.

Proposed approach

This study introduces a two-stage hybrid architecture that combines Parameter-Efficient Fine-Tuning (PEFT) and stacking ensembles to perform sentiment analysis efficiently. Our main goal is to reduce the computational bottleneck of training Large Language Models (LLMs) while maintaining high classification performance. We structure our approach into four main phases:

Phase 1: Base Model Adaptation: We utilize pre-trained **RoBERTa** and **DeBERTa** models as our base learners. Instead of fully fine-tuning them which is highly computationally expensive we freeze their original weights. We then insert Low-Rank Adaptation (**LoRA**) matrices into their attention layers. This allows us to train the models by updating only a very small fraction of the parameters, significantly saving memory and time.

Phase 2: Out-of-Fold (OOF) Predictions: To prevent data leakage, we are considering implementing a 5-fold cross-validation strategy. Our base models will be trained on four folds, and probability predictions will be generated on the hidden fifth fold, thus providing clean data for the next phase.

Phase 3: Meta-Learner Training: To leverage the strengths of both the RoBERTa and DeBERTa models simultaneously, we combine the OOF predictions of both into a single feature vector. We will then feed these combined features to a meta-learning algorithm that we will define later, so that it learns how to weight the predictions of these two fundamental models for the decision-making process.

Phase 4: Inference: During the testing phase, a new review passes through both LoRA-adapted base models to generate initial probability scores. The meta-learner then processes these combined scores to output the final positive or negative sentiment prediction.

Baseline Methods

To properly evaluate the effectiveness of our proposed method, we will compare it against three baselines:

Baseline 1: TF-IDF with SVM: A traditional machine learning approach using Term Frequency-Inverse Document Frequency (TF-IDF) and a Support Vector Machine. This acts as our lower-bound reference, demonstrating the performance gap between classical sparse representations and contextual transformer models.

Baseline 2: Full Fine-Tuned RoBERTa: We will fully fine-tune a standard RoBERTa model to serve as our upper bound for computational cost. Comparing our approach to this baseline will

clearly show how much training time and VRAM we save using LoRA, and whether we sacrifice any accuracy in the process.

Baseline 3: Single PEFT Model (RoBERTa + LoRA): To prove that the stacking ensemble actually adds value, we will compare our final two-model ensemble against a single RoBERTa model trained only with LoRA. This ablation study will help us measure the specific performance gain achieved by combining different architectures.

Dataset

The dataset used in this project's named "IMDB Dataset of 50k Movie Reviews" and can be reached from the following website <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews> (66MB). It contains 49582 unique instances and only one feature represented by the entire review written by a user. The output is of categorical kind and represents the sentiment expected from the review, so the only two possible categories are positive and negative.

Description of the dataset(s) to be used, including source and size if available.

Evaluation strategy

A first metric to classify the quality of our fine tuned model is the accuracy, measuring the correct predictions out of all predictions. As this metric needs to be handled carefully, other metrics such as Precision, Recall and F1-Score should be evaluated as well to put the accuracy in the right context.

Expected challenges

While the proposed architecture offers significant advantages, several technical challenges are anticipated during the implementation phase. Acknowledging and addressing these challenges is crucial for the robust execution of the project:

Handling Long-Text Token Limits: In the architecture we will be using, RoBERTa and DeBERTa can use a maximum of 512 tokens. The comments within the IMDb dataset contain data that exceeds this limit; therefore, if the data cannot be retrieved from the beginning or end, an incorrect conclusion may be reached.

Data Leakage and Overfitting Risk in Stacking: If the Level 1 Meta-Learner is trained and validated using the same data we used to train our Level 0 models, and the resulting probability estimates are used again, an overfitting scenario will likely occur. To resolve this, we will adhere to the Out-of-Fold (OOF) estimation generation method.

LoRA Hyperparameter Optimization Overhead: Achieving the optimal balance between performance and parameter efficiency relies heavily on tuning LoRA-specific hyperparameters, particularly the rank dimension (r), the scaling factor (α), and dropout rates. Since this tuning is inherently an empirical, trial-and-error process, it presents a risk of unforeseen computational resource and time consumption.

GPU Memory Management (OOM Constraints): Although Parameter-Efficient Fine-Tuning (PEFT) drastically reduces the number of *trainable* parameters, the underlying base models (especially DeBERTa) remain architecturally heavy and demand substantial VRAM for forward passes and activations. Loading these models during the ensemble training and inference phases poses a persistent risk of Out-of-Memory (OOM) errors. Meticulous hardware monitoring, careful optimization of training batch sizes, and the potential implementation of gradient accumulation techniques will be essential.